

Title: Impact of X-waiver policies on buprenorphine prescribing for opioid use disorder in US, 2019-2025

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Objectives

The primary objective evaluated the impact of X-waiver policies on the number and period prevalence of individuals with OUD who were prescribed buprenorphine from 2019 to 2025.

The secondary objective evaluated the impact of X-waiver policies on buprenorphine prescribing across social vulnerability index (SVI) quantiles from 2019 to 2025.

Methods

Study design

An interrupted time series analysis (ITSA) was used to evaluate the trends and level changes associated with X-waiver policies on buprenorphine prescribing for adults with opioid use disorder (OUD) in the United States (US). Two X-waiver policy interventions were evaluated with the X-waiver relaxation policy occurring in April 2021 (time = 29) and the X-waiver elimination policy occurring in January 2023 (time = 50). The period before the first X-waiver policy change is labelled “Segment 1,” the period after the first X-waiver policy and before the second X-waiver policy is labelled “Segment 2,” and the period after the second X-waiver policy is labelled “Segment 3.” We used 30-day time intervals instead of monthly time intervals to establish a consistent replicable dataset. Consequently, our study’s time horizon ended on 25 December 2025. The online supplement provides a visualization of the ITSA framework ([see online Supplement A](#)).

Sample and Social Vulnerability Index

Adults (≥ 18 years old) with a diagnosis of opioid use disorder were included for analysis. Individuals with opioid use disorder were categorized according to their social vulnerability index (SVI) quantiles. SVI index was based on data from the Centers for Disease Control and Prevention (CDC).^{1,2} Epic Cosmos assigned SVI score to individuals using Federal Information Processing Standards (FIPS) codes. Individuals are considered least socially vulnerable if they are in the first quantile (SVI = 1) and most socially vulnerable if they are in the fourth quantile (SVI = 4).

Data source

Epic Cosmos was used to gather data on the number of buprenorphine prescriptions prescribed to unique individuals with opioid use disorder between 01 January 2019 to 25 December 2025. Epic Cosmos is a collaborative network of health systems that aggregates data from institutions that use the Epic Electronic Health Record System for approximately more than 300 million unique individuals.^{3,4}

Opioid use disorder diagnosis was determined based on ICD10 codes.

Outcome variable

Buprenorphine prescriptions dispensed were determined using Epic Cosmos. Dispensed buprenorphine prescriptions are based on Epic's Medication (ERX) master file, which is constructed using from third-party medication databases such as First DataBank and Medi-Span in the United States.

We opted to measure the number of individuals with OUD diagnosis prescribed buprenorphine and the period prevalence of individuals with OUD prescribed buprenorphine per 1000 persons due to the changing number of individuals with OUD in the US population.

The period prevalence (number of individuals with OUD prescribed buprenorphine per 1000 persons) was estimated as:

$$\text{Period prevalence} = \left(\frac{\text{Number of individuals with OUD prescribed buprenorphine}}{\text{Number of individuals with OUD}} \right) * 1000$$

Statistical analysis

Descriptive analyses on the number of individuals with OUD prescribed buprenorphine and the number of individuals with a diagnosis of OUD was performed from 2019 to 2025.

In the primary analyses, two ITSAs were constructed using (1) the number of individuals with OUD who were prescribed buprenorphine and (2) period prevalence of individuals with OUD who were prescribe buprenorphine as outcomes.

For the first outcome, a linear mixed effects model with a random intercept was used to evaluate the number of buprenorphine prescriptions prescribed during the study time horizon and their associations with X-waiver policies changes.

For the second outcome, a negative binomial model was used to evaluate the period prevalence of buprenorphine prescriptions prescribed during the study time horizon and their associations with X-waiver policies changes. We used the log of the population size of adults with opioid use disorder as the offset term in the model.

For both models, the mean monthly changes (slopes) for each period segments and the level changes at each period when the X-waiver policies were implemented were estimated. For the primary analyses, differences between slopes between each segment were estimated. Robust standard errors were estimated for all measures except for differences in slopes, which were estimated using non-parametric bootstrap with 1000 replications. Stata does not allow for estimation of standard errors in terms of the period prevalence; hence, bootstrap methods were used. Results are presented as the mean change along with their corresponding 95% confidence intervals (CI).

For the secondary analyses, social vulnerability index (SVI) quantiles were used as the grouping variable to evaluate the impact of X-waiver policy changes on buprenorphine prescribing among

adults with opioid use disorder. Interaction terms between the SVI quantiles (grouping variable) and the period when the X-waiver policy was relaxed (time = 29) and eliminated (time = 50) were used to compare the differences in buprenorphine prescribing between individuals in different SVI quantiles. Similar approaches were applied to the SVI quantiles analyses as in the primary analyses.

Statistical significance was defined as a two-tailed $\alpha < 0.05$. All analyses were performed using Stata SE 18 (StataCorp, LLC, College Station, TX).

Results

By December 2025, 1,213,349 individuals had a diagnosis of OUD compared to 444,739 in January 2019—a 173% relative increase (**see online Supplement B**). The total number of individuals with OUD who received buprenorphine increased from 71,508 in January 2019 to 319,371 in December 2025—a 347% relative increase.

Primary analyses

In the primary analyses, there was a significant increase in the number of individuals with OUD who received buprenorphine from Segment 1 to Segment 2 and from Segment 2 to Segment 3 (**Table 1, Figure 1A**). Between Segment 1 to Segment 2, the number of individuals with OUD who received buprenorphine increased by 246 prescriptions (95% CI: 135, 357). Between Segment 2 to Segment 3, the number of individuals with OUD who received buprenorphine increased by 515 prescriptions (95% CI: 413, 617). Immediately after the X-waiver relaxation policy was implemented, there was a significant increase in the number of individuals with OUD who received buprenorphine (+1612; 95% CI: 133, 3091). Similarly, immediately after the X-waiver elimination policy was implemented, there was a significant increase in the number of individuals with OUD who received buprenorphine (+1454; 95% CI: 13, 2894).

Conversely, there was a significant decrease in the period prevalence of individuals with OUD who received buprenorphine from Segment 1 to Segment 2 and from Segment 2 to Segment 3 (**Table 1, Figure 1B**). Between Segment 1 to Segment 2, the number of individuals with OUD who received buprenorphine decreased by 1.60 prescriptions per 1000 persons (95% CI: (-1.68, -1.51)). Between Segment 2 to Segment 3, the number of individuals with OUD who received buprenorphine decreased by 0.27 prescriptions per 1000 persons (95% CI: -0.34, -0.20). Immediately after the X-waiver relaxation policy was implemented, there was a significant decrease in the period prevalence of individuals with OUD who received buprenorphine (-2.88; 95% CI: -4.34, -1.42). However, immediately after the X-waiver elimination policy was implemented, there was a significant increase in the period prevalence of individuals with OUD who received buprenorphine (+1.21; 95% CI: 0.27, 2.15).

Secondary analyses

In the secondary analyses, the number of individuals with buprenorphine was greater among those with greater social vulnerability compared to those with lesser social vulnerability (**Table 2, Figures 2A and 2B**). Significant increases in the number of individuals with OUD who received buprenorphine were observed for all SVI quantiles (**Figure 2A**). Among the least vulnerable (SVI = 1) the number of individuals with buprenorphine increased from a rate of 307

per month (in Segment 1) to a rate of 361 per month (in Segment 2), a relative increase of 17.6% $[(361 - 307) / 307]$. Similarly, among the most vulnerable (SVI = 4), the number of individuals with buprenorphine increased from a rate of 486 per month (in Segment 1) to 526 per month (in Segment 2), a relative increase 8.3% $[(526 - 486) / 486]$. This pattern was observed when individuals transitioned from Segment 2 to Segment 3 (**Table 2**); there was a 7.8% relative increase $[(389 - 361) / 361]$ among the least vulnerable (SVI = 1), and a 41.3% relative increase $[(743 - 526) / 526]$ among the most vulnerable (SVI = 4).

However, when reporting on period prevalence outcomes, the trends were mostly reversed. Significant decreases in the period prevalence of individuals with OUD who received buprenorphine were observed for most SVI quantiles (**Figure 2B**). Among the least vulnerable (SVI = 1) the period prevalence of individuals with buprenorphine decreased from a monthly rate of 3.62 per 1000 persons (in Segment 1) to a monthly rate of 1.63 per 1000 persons (in Segment 2), a relative decrease of 55.0% $[(1.63 - 3.62) / 3.62]$. Similarly, among the most vulnerable (SVI = 4), the period prevalence of individuals with buprenorphine decreased from a monthly rate of 4.01 per 1000 persons (in Segment 1) to 1.44 per 1000 persons (in Segment 2), a relative decrease 64.1% $[(1.47 - 4.01) / 4.01]$. Between Segment 2 and Segment 3, there was a 41.7% relative decrease in the monthly rate among the least vulnerable individuals (SVI = 1) with OUD (**Table 2**). Individuals in SVI quantiles 2 and 3 followed similar patterns as individuals in SVI quantile 1. However, among the most vulnerable (SVI = 4), there was a 18.4% relative increase $[(1.74 - 1.47) / 1.47]$ in the monthly rate when transitioning from Segment 2 to Segment 3.

Discussion

The impact of the X-waiver policies has had mixed effects on buprenorphine prescribing. When reporting the impact of these X-waiver policies on the number of individuals with OUD who were prescribed buprenorphine, the trends indicated that there was a significant increase. However, when reporting on the period prevalence of individuals with OUD who were prescribed buprenorphine, the trends were mostly negative. These apparent differences can be explained by the type of outcomes used in our analysis.

Period prevalence takes into consideration the population at risk, which in our case were those individuals with OUD; whereas simply relying on the number of individuals with OUD who were prescribed buprenorphine does not capture this changing trend in the population at risk. In our descriptive analysis, we reported that the population of individuals with OUD increased at a greater rate than individuals who were prescribed buprenorphine, which has had an impact the observed positive trends of the numerator. Although the number of individuals with OUD who were prescribed buprenorphine increased, this pattern reverses when the total population at risk was incorporated. Decision makers can use both outcomes to influence policy; however, caution should be exercised when the denominator undergoes substantial changes across time.

Additionally, the opioid crisis has resulted in a large number of new OUD diagnosis, which has overwhelmed the capacity of public health efforts to improve access to essential treatments such as buprenorphine.⁵⁻⁷ These findings highlight the challenges in meeting the high demand of the OUD population for buprenorphine. As long as the OUD population continues to increase, any positive trends in the number of individuals with OUD who are prescribed buprenorphine will be attenuated. Therefore, it is essential that decision makers use an outcome measure that captures both the change in the number of individuals with OUD who were prescribed buprenorphine and the prevalence individuals with OUD who were prescribed buprenorphine for resource planning and understanding the burden of the disease.

In our analyses, we identified certain patterns in buprenorphine prescribing across SVI quantiles. When viewing the number of individuals with OUD who were prescribed buprenorphine, we observed that those in the lowest social vulnerability index (SVI = 1) has the least amount of buprenorphine compared to those in the highest social vulnerability index (SVI = 4). This was counter to our expectations that individuals in a socially vulnerable environment would have lower opportunities for access to medication treatment for opioid use disorder. Yang and colleagues reported that among older adults (≥ 65 years), the number of individuals with OUD is greater in counties with high social vulnerability compared to counties with low social vulnerability.⁸ Similarly, Joudrey and colleagues reported that counties with greater social vulnerability had limited access to buprenorphine and other medications for OUD treatment.⁹ Lastly, other community-level factors such as high provider density and high mental health service availability interact with SVI to improve buprenorphine retention.¹⁰

Previous studies are mixed when it comes to the impact of the X-waiver policies on buprenorphine prescribing. Stone and colleagues reported that the X-waiver elimination was associated with increased clinicians prescribing buprenorphine but an overall decrease in patients receiving buprenorphine.¹¹ Similarly, Chua, Bohnert, and Nguyen reported a significant increase in the number of buprenorphine prescribers, but a limited impact on buprenorphine

prescriptions.¹² Conversely, Tuan and colleagues reported that elimination of the X-waiver was associated with a 14% increase in the odds of buprenorphine initiation after a new OUD diagnosis.¹³ We speculate that the differences in these findings may depend on the type of patients receiving buprenorphine and the specialties of their providers. For instance, Stone and colleagues reported that there was an overall decrease in buprenorphine prescribing by all physician groups after the X-waiver except for behavioral health physicians.¹¹

Limitations

This study has several limitations. First, the number and cumulative prevalence of individuals with OUD who were prescribed buprenorphine were based on a single electronic health record system that may not be representative of the whole US population. Epic Cosmos only captures data on patients who engaged with healthcare systems that use its Epic Electronic Health Record System. Thus, it does not capture other patients outside this platform, and any findings may not be reflective of the general US population. Second, the data are an aggregate of individuals with OUD and do not include patient-level characteristics. Consequently, we were unable to control for patient-level characteristics in our models, which introduces potential confounding issues. Additionally, since the data are aggregated for the US, ecological fallacy could be present.^{14,15} To address this, we grouped the data into SVI quantiles to observe any difference in community-level vulnerabilities as a secondary aim. However, this strategy only allows us to stratify the findings across SVI quantiles and does not adjust for potential confounding. Lastly, OUD diagnosis is challenging to diagnose and could lead to misclassification bias. Epic Cosmos used ICD10 diagnostic codes to capture OUD diagnosis, which has been reported to be insufficient in properly identifying OUD and could result in potential misclassification.¹⁶

Conclusions

Overall, the X-waiver policies appeared to have the intended effect of increasing buprenorphine prescribing in terms of raw numbers, but the change in the period prevalence of individuals with OUD who received buprenorphine was not a great in the periods after the X-waiver policies compared to before. Selection of outcomes can influence interpretations of findings; thus, it is recommended that presentation of findings include all outcomes.

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